FOR EDITH · PAEDIATRICS & CHILD HEALTH JUNIOR CLERKSHIP

Acute Rheumatic Fever & Rheumatic Heart Disease

Tutorial | Week 4 · Small babies, sugar & the heart | Paediatrics | Dr Aliku

Acute rheumatic fever is an autoimmune reaction 2–4 weeks after group A streptococcal sore throat that inflames joints, brain, skin and the heart; Uganda is a high-risk setting, so use the 2015 revised Jones criteria with high-risk thresholds, treat the acute illness and any heart failure, and above all give benzathine penicillin every 4 weeks for years, because only secondary prophylaxis stops rheumatic heart disease progressing.

Must know before the ward round

1. Revised Jones criteria (AHA 2015, still current, endorsed by WHO 2024): initial ARF = evidence of preceding GAS infection plus 2 major, or 1 major + 2 minor. Recurrence = 2 major, 1 major + 2 minor, or 3 minor.
2. Major (J♥NES): Joints (arthritis), Carditis (clinical or subclinical on echo), Nodules (subcutaneous), Erythema marginatum, Sydenham chorea.
3. In high-risk populations like Uganda, MONOarthritis or polyarthralgia can be a major criterion. Minor criteria are monoarthralgia, fever ≥38.0 °C, ESR ≥30 mm/h and/or CRP ≥3.0 mg/dL (30 mg/L), and a prolonged PR interval.
4. Sydenham chorea (and indolent carditis) alone is enough to diagnose ARF, with no other criteria or streptococcal evidence needed.
5. Eradicate GAS at diagnosis: benzathine penicillin G (BPG) IM single dose, 0.6 MU if under 30 kg and 1.2 MU if 30 kg or more (UCG 2023; AHA uses 27 kg), or penicillin V/amoxicillin for 10 days. The same injection starts secondary prophylaxis.
6. Arthritis: aspirin 80–100 mg/kg/day in 3–4 doses (UCG 2023) until symptoms settle, then taper; naproxen 10–20 mg/kg/day in 2 doses (max 1 g/day) is a safer alternative. Arthritis should improve dramatically within 24–72 hours; if not, rethink the diagnosis.
7. Heart failure from carditis: furosemide 1 mg/kg (WHO), spironolactone, an ACE inhibitor (enalapril 0.1 mg/kg/day, titrated) and fluid/salt restriction. Prednisolone 1–2 mg/kg/day is used for severe carditis, though evidence is weak.
8. Secondary prophylaxis: BPG IM every 4 weeks (every 3 weeks if breakthrough or high-risk). Oral penicillin V 250 mg twice daily if injections are impossible; erythromycin if truly allergic.
9. Duration (UCG 2023): no carditis = 5 years or until age 18–21. Carditis without residual disease = 10 years or until age 25. Residual valve disease = until age 40–45 or for life. (AHA 2020: whichever is longer of 5 y/age 21, 10 y/age 21, 10 y/age 40.)
10. GOAL trial (Uganda, NEJM 2022): BPG every 4 weeks in children with latent (echo-detected) RHD cut 2-year progression from 8.2% to 0.8%. WHO 2024 now recommends prophylaxis for screen-detected RHD in children.
11. RHD most often damages the mitral valve (regurgitation in children, then stenosis), then the aortic valve. Apical pansystolic murmur radiating to the axilla = MR; Carey Coombs mid-diastolic murmur = acute mitral valvulitis.
12. WHO 2024: penicillin (IM or oral) is first line for GAS pharyngitis. In high-risk settings without testing, treat clinically suspected GAS sore throat in children with antibiotics.

1 In a nutshell
2 Definitions & classification
3 Epidemiology & Uganda context
4 Pathophysiology
5 Clinical acumen
6 Clinical workup
7 Differential diagnosis
8 Investigations
9 Management
10 Complications
11 Ward-round & exam pearls
12 Mnemonics & memory aids
13 High-yield summary

In a nutshell

Acute rheumatic fever (ARF) is an **autoimmune inflammatory illness** that follows an untreated **group A *Streptococcus* (GAS, *Streptococcus pyogenes*) throat infection** (and possibly skin infection) by about **2–4 weeks**. It mainly affects **children aged 5–15 years**. It causes **migratory arthritis, carditis, chorea, skin rash and nodules**. The joints, brain and skin recover completely, but the **heart valves may scar permanently**. This is **rheumatic heart disease (RHD)**.

RHD is the **most common acquired heart disease in people under 25** worldwide (WHO: about **55 million affected, 360,000 deaths a year**), almost all in low- and middle-income countries. Uganda is a **high-risk setting**. Many Ugandan children are first diagnosed when they already have **heart failure from severe mitral or aortic valve disease**.

The disease is **preventable at three levels**:

1. **Primordial**: housing, less crowding, nutrition.
2. **Primary**: treat GAS sore throats with penicillin.
3. **Secondary**: **benzathine penicillin G (BPG) injections every 4 weeks** for years after ARF or RHD diagnosis. This is the single most important intervention you will prescribe.

> **CLINICAL PEARL**
> The one thing never to forget: **ARF "licks the joints but bites the heart."** The arthritis is dramatic and goes away; the carditis is often silent and stays. Every child with ARF or RHD needs an **echocardiogram** and **monthly BPG**, and a missed injection is a chance for a recurrence that damages the valves further.

Definitions & classification

The 2015 revised Jones criteria (AHA; endorsed by WHO 2024)

First decide whether the child comes from a **low-risk** population (ARF incidence ≤2 per 100,000 school-aged children per year, or all-age RHD prevalence ≤1 per 1,000) or a **moderate/high-risk** population (everyone else). **Ugandan children are moderate/high risk.**

	Low-risk populations	Moderate/high-risk (Uganda)
Major	Carditis (clinical and/or subclinical), **polyarthritis only**, chorea, erythema marginatum, subcutaneous nodules	Carditis (clinical and/or subclinical), **monoarthritis or polyarthritis, and/or polyarthralgia** (if other causes excluded), chorea, erythema marginatum, subcutaneous nodules
Minor	Polyarthralgia, fever **≥38.5 °C**, **ESR ≥60 mm/h** and/or CRP ≥3.0 mg/dL, prolonged PR (age-adjusted, unless carditis is a major criterion)	**Monoarthralgia**, fever **≥38.0 °C**, **ESR ≥30 mm/h** and/or **CRP ≥3.0 mg/dL (30 mg/L)**, prolonged PR (age-adjusted, unless carditis is a major criterion)

Diagnosis

Situation	Requirement
Initial ARF	Evidence of preceding GAS infection + **2 major** or **1 major + 2 minor**
Recurrent ARF	Evidence of preceding GAS infection + **2 major**, **1 major + 2 minor**, or **3 minor**
Sydenham chorea or **indolent carditis**	Can be diagnosed **alone**, without other criteria or GAS evidence (they appear months after the infection)

Evidence of preceding GAS infection: raised or rising **antistreptolysin O (ASO)** titre or **anti-DNase B**; positive **throat culture**; positive **rapid GAS antigen test**; recent scarlet fever (WHO Pocket Book).

> UGANDA CONTEXT
> **UCG 2023** prints a simplified Jones table: evidence of strep infection + 2 major or 1 major + 2 minor (polyarthritis listed as major, ESR/CRP and PR prolongation as minor). For exams and the ward, use the **2015 high-risk version**, which the **WHO 2024 guideline** endorses ("the Jones criteria should be used"). It lets you count **monoarthritis and polyarthralgia** as major, which matters because many Ugandan children have already had NSAIDs or herbal remedies before arthritis becomes "poly".

Subclinical carditis and latent RHD

- **Subclinical carditis**: no murmur, but **pathological mitral or aortic regurgitation on Doppler echocardiography**. Counts as carditis (major criterion) since 2015.
- **Latent RHD**: valve changes found by **echo screening** in an asymptomatic child who never had recognised ARF. The **World Heart Federation (WHF) criteria** (2012, updated 2023) define it, and **WHO 2024** recommends prophylaxis for children meeting minimum criteria.
- **Pathological MR on Doppler (WHF)**: seen in **2 views**, jet length **≥2 cm** in at least one view, peak velocity **≥3 m/s**, and **pansystolic** in at least one envelope. **Pathological AR**: 2 views, jet **≥1 cm**, velocity ≥3 m/s, **pandiastolic**.

Heart failure severity

- **Infants and young children**: modified **Ross** classification (feeding, sweating, tachypnoea, growth).
- **Older children**: **NYHA class I–IV** (I: no limitation; II: symptoms on ordinary activity; III: symptoms on less than ordinary activity; IV: symptoms at rest).

Epidemiology & Uganda context

- **Age**: first episode usually at **5–15 years** (UCG: 3–15). Rare under 3. Recurrences can continue into adulthood.
- **Risk factors**: **overcrowding** (school dormitories, shared beds), poverty, poor access to treatment of sore throat, undernutrition, genetic susceptibility (HLA types; familial clustering), female sex for RHD.
- **Uganda**: echocardiographic school screening studies (Kampala, Gulu, Lira and others) found latent RHD in roughly **1.5–3% of school-age children** (VERIFY district figures). Most of these children had never been diagnosed with ARF, because ARF often presents as "joint pains" treated at drug shops.
- **Services**: the **Uganda Heart Institute (UHI) at Mulago** is the national referral centre for echo, catheter interventions (e.g. balloon mitral valvotomy) and open-heart surgery. **RHD clinics** run at several regional referral hospitals (e.g. **Lira, Gulu, Mbarara**; VERIFY current list), linked to a national **RHD registry**. Many children still need surgery abroad, funded by charities.
- **BPG supply**: global stock-outs of benzathine penicillin have affected Uganda. Families travel monthly for injections, so **adherence** is a major problem.
- **Pregnancy**: girls with RHD later face high risks in pregnancy (see the obs topic on cardiac disease in pregnancy). WHO 2024 supports **echo screening in antenatal care** in high-risk settings.

Pathophysiology

From sore throat to scarred valve

1. **GAS pharyngitis** (or possibly impetigo) in a genetically susceptible child.
2. **Molecular mimicry**: antibodies and T cells against streptococcal **M protein** and **N-acetylglucosamine** (cell wall carbohydrate) **cross-react** with human proteins: **cardiac myosin**, **laminin and keratin on valve endothelium**, **joint synovium**, and **basal ganglia neurons** (lysoganglioside, tubulin).
3. After a **latent period of 2–4 weeks** (1–5 weeks), inflammation appears in joints, heart, skin and brain. **Chorea** has a longer latency (**1–6 months**), so ASO may have normalised by then.
4. **Carditis is a pancarditis** (endocardium, myocardium, pericardium), but **valvulitis** causes the lasting damage. **Aschoff bodies** (granulomas with Anitschkow "caterpillar" cells) are the histological hallmark.
5. **Repeated attacks** cause more scarring: leaflet thickening, **commissural fusion**, chordal shortening and fusion, leading to **regurgitation** (early, in children) and then **stenosis** (later, but can be severe in African adolescents: "juvenile mitral stenosis").

Valve involvement: **mitral (~65–70%)** > **mitral + aortic (~25%)** > aortic alone > tricuspid (rarely isolated, usually functional from pulmonary hypertension). Pulmonary valve almost never.

Why the heart fails

- **Acute carditis**: severe MR (valvulitis + annular dilatation + chordal elongation) → volume overload → left atrial and pulmonary venous congestion → **pulmonary oedema**. Myocarditis contributes less than once thought.
- **Chronic RHD**: MR → left ventricular dilatation; MS → raised left atrial pressure → **pulmonary hypertension**, **atrial fibrillation**, right heart failure; AR → LV volume overload, wide pulse pressure.
- **Precipitants of decompensation**: **ARF recurrence**, infection (pneumonia, **infective endocarditis**), anaemia, AF, pregnancy, stopping medication.

Clinical acumen

Focused history

Ask about	Why it matters
Sore throat 2–4 weeks ago, treatment received, scarlet fever, skin sores/scabies	Preceding GAS; many children recall nothing (only ~⅔ do)
Joint pain: which joints, **migratory** (one gets better as another flares), swelling, **exquisitely painful**, prior NSAIDs/aspirin	Classic ARF arthritis is migratory, large joints, very painful, and responds dramatically to NSAIDs (which also mask the migration)
Fever	Minor criterion
Breathlessness, orthopnoea, cough, chest pain, palpitations, swelling of feet/face, reduced exercise tolerance	Carditis, heart failure, pericarditis, AF
Clumsiness, dropping things, writing deteriorating, emotional lability, "restlessness" noticed by teachers	**Sydenham chorea**
Rash, painless lumps	Erythema marginatum, nodules
Previous ARF, known RHD, BPG injections (when last, how many missed)	Recurrence; adherence
Penicillin allergy history (what exactly happened)	Most "allergies" are not true allergy; WHO 2024 advice on test doses

Ask about	Why it matters
Family history of RHD, household crowding, school (boarding?)	Risk and household screening
Feeding, weight loss, growth, immunisation, HIV status	Nutrition and comorbidity; heart failure causes poor growth
Menarche and sexual activity (adolescent girls)	Pregnancy risk with RHD; contraception counselling; warfarin teratogenicity

Examination

- **General**: comfortable or distressed, **posture** (sitting up = orthopnoea), pallor (anaemia worsens failure), growth (weight, height, Z-scores).
- **Joints**: swelling, warmth, redness, **severe tenderness out of proportion**, limited movement; usually knees, ankles, elbows, wrists.
- **Skin**: **erythema marginatum** (pink rings with pale centres on the trunk and proximal limbs, non-itchy, evanescent; hard to see on dark skin), **subcutaneous nodules** (painless, pea-sized, over extensor surfaces, occiput and spine; usually with carditis).
- **Chorea**: involuntary, purposeless, jerky movements (face, hands); **milkmaid's grip** (squeezes and releases rhythmically), **darting tongue** ("jack-in-the-box"), **pronator sign** (arms held up turn palms outward), **spooning** of outstretched hands, hypotonia, emotional lability. Worse with stress, absent in sleep.
- **Cardiovascular**: pulse rate (**tachycardia out of proportion to fever**, persistent in sleep), rhythm (AF in chronic MS), volume (**collapsing** in AR), BP (wide pulse pressure), JVP, **apex position and character**, heaves, thrills; **murmurs** (see below), **pericardial rub**, gallop; liver size, oedema, basal crackles.

Murmur	Lesion	Where and how
Apical pansystolic, radiating to the axilla	**Mitral regurgitation** (commonest in children)	Diaphragm; louder lying on the left side
Apical mid-diastolic (Carey Coombs)	**Acute mitral valvulitis** (high flow across an inflamed valve)	Bell, left lateral position; no opening snap
Apical low-pitched mid-diastolic rumble with presystolic accentuation, loud S1, opening snap	**Mitral stenosis** (chronic)	Bell, left lateral position, after exercise
Early diastolic decrescendo, left sternal edge	**Aortic regurgitation**	Diaphragm, sitting forward in held expiration
Pericardial friction rub	Pericarditis	Scratchy, varies with position

Red flags

RED FLAG

- **Heart failure**: tachypnoea, orthopnoea, tachycardia, gallop, hepatomegaly, raised JVP, oedema.
- **New AF or embolic event** (stroke): anticoagulation and urgent cardiology review.
- **Fever + changing murmur + splinter haemorrhages/haematuria/splenomegaly** in a child with RHD: **infective endocarditis**. Take 3 blood cultures before antibiotics.
- **Syncope, chest pain on exertion**: severe aortic stenosis or pulmonary hypertension.
- **Severe MS with haemoptysis or pulmonary oedema**: urgent referral to UHI (balloon valvotomy or surgery).
- **Pregnancy in an adolescent with RHD**: high-risk; refer.

Thinking like a clinician

- **"Joint pains" in a Ugandan school child**: think ARF, but also **sickle cell crisis**, **septic arthritis** (single hot joint, very unwell, needs aspiration), **reactive arthritis**, **juvenile idiopathic arthritis**, **leukaemia** (bone pain, cytopenias), **post-streptococcal reactive arthritis** (non-migratory, poor response to NSAIDs). A dramatic response to aspirin within 1–3 days supports ARF.
- **Arthritis that does not respond** to NSAIDs within 72 hours → rethink.
- **Tachycardia at rest and in sleep, out of proportion to fever** → think carditis even before a murmur appears.
- **Every ARF patient needs an echo**: about a third or more of "no carditis" cases have subclinical carditis, which changes prophylaxis duration.
- **A girl with clumsiness and emotional changes** whose teacher thinks she is "playing the fool" may have **Sydenham chorea**. ASO may be normal by then.
- **Infective endocarditis vs ARF recurrence** in a known RHD child with fever: both give fever, raised CRP and a changing murmur. Always do **blood cultures** before antibiotics.
- **Student mistake**: forgetting to **give BPG at diagnosis** and to **book the next injection**. The prescription is incomplete without a prophylaxis card and a date.

Clinical workup

Step 0: First 5 minutes

1. **ABC**: respiratory distress, SpO_2, sit the child up if orthopnoeic, **oxygen** if SpO_2 <90% (or <94% with emergency signs).
2. **Circulation**: HR (age-adjusted), rhythm, BP, CRT, liver size, JVP.
3. **Signs of heart failure** → start furosemide 1 mg/kg IV (WHO), restrict IV fluids.
4. **Temperature, Hb (HemoCue)** (anaemia worsens failure), glucose if unwell.
5. **ECG** if irregular pulse or very fast (AF, heart block).
6. If fever + known RHD + new embolic signs or splinters: **blood cultures × 3 before antibiotics** (endocarditis).

Step 1: Focused history

- Sore throat or skin sores 2–6 weeks ago; treatment received.
- Joint pattern, migration, response to painkillers.
- Cardiac symptoms; exercise tolerance (NYHA).
- Chorea symptoms (school performance, handwriting).
- Previous ARF/RHD, **BPG card** (dates, missed doses).
- Allergy history; current medicines (aspirin, warfarin).
- Growth, nutrition, immunisation, HIV status, menstrual history.

Step 2: Focused examination

- Vitals (HR, RR, BP, SpO_2, temperature), weight and height with Z-scores.
- Joints (each large joint), skin (rash, nodules), neurological (chorea signs).
- Full cardiovascular examination (see the paediatric cardiovascular exam topic): apex, heaves, thrills, **each murmur by timing, site, grade, radiation**, rub, signs of failure.
- Look for endocarditis stigmata: clubbing, splinters, Osler nodes, Janeway lesions, splenomegaly, microscopic haematuria, Roth spots.

Step 3: Bedside tests

Test	Use
ECG	PR prolongation (minor criterion), heart block, AF, LVH/LA enlargement, low voltages (pericardial effusion)
Urine dipstick	Haematuria in endocarditis; protein (PSGN can coexist)
HemoCue Hb	Anaemia
Rapid GAS antigen test (if available)	Current GAS pharyngitis
Handheld or point-of-care echo	WHO 2024 supports **handheld echocardiography** for diagnosing RF carditis and RHD where standard echo is unavailable

Age-adjusted upper limit of the PR interval: 3–12 years **0.16 s**; 12–16 years **0.18 s**; ≥17 years **0.20 s**.

Step 4: Laboratory & imaging

Test	When	Expected / abnormal finding	How it changes management
ESR, CRP	Every suspected ARF; repeat every 1–2 weeks	ESR ≥30 mm/h or CRP ≥30 mg/L (high-risk thresholds); falling with treatment	Minor criterion; guides duration of anti-inflammatory therapy and bed rest. Most hospitals
ASO titre	Every suspected ARF	Raised (above age-specific upper limit; often >200–300 IU/mL in school children) or **rising on repeat after 10–14 days**	Evidence of GAS. Peaks 3–6 weeks after infection; **anti-DNase B** stays raised longer (useful after skin infection or late presentation). ASO at hospitals; anti-DNase B at referral labs
Throat swab culture	Before antibiotics	GAS (positive in only ~10–25% by the time ARF appears)	Evidence of GAS. RRH/NRH
FBC	All	Anaemia (of inflammation), leucocytosis	Treat anaemia; exclude leukaemia (cytopenias, blasts)
Echocardiography with Doppler	**Every suspected ARF and all known RHD**	Pathological MR/AR (WHF), valve thickening, chordal changes, pericardial effusion, LV dilatation and function, MS gradient, pulmonary pressure	Confirms carditis (including subclinical), stages RHD, decides prophylaxis duration and referral for intervention. UHI, RRHs with cardiac clinics, private centres; handheld echo in programmes
Chest X-ray	Heart failure, chest symptoms	Cardiomegaly, pulmonary venous congestion, LA enlargement (double right heart border, splayed carina), effusion	Severity; alternative diagnoses
Blood cultures (× 3)	Fever in known RHD, suspected endocarditis	Viridans streptococci, *Staph aureus*, HACEK	Endocarditis: 4–6 weeks IV antibiotics
U&E, creatinine	Before ACE inhibitors and diuretics, and during	Rising creatinine, low or high K^+	Dose adjustments
INR	On warfarin	Target 2–3 (AF), 2.5–3.5 (mechanical mitral valve)	Dose adjustment; limited availability outside referral centres

Step 5: Putting it together

Model one-liner: "A 12-year-old girl from a boarding school in Lira with **migratory polyarthritis** of the knees and ankles for 10 days, fever 38.4 °C, a **new apical pansystolic murmur** radiating to the axilla, **ESR 78**, **ASO 640 IU/mL** and echo showing **moderate pathological MR**: **initial acute rheumatic fever with carditis** (2 major + 2 minor criteria with GAS evidence)."

Worked example: applying Jones (high-risk)

Child	Findings	Major	Minor	GAS evidence	Verdict
A, 9 y	Swollen painful right knee (monoarthritis), fever 38.2, CRP 60 mg/L, ASO high	Monoarthritis (1)	Fever, CRP (2)	Yes	**Initial ARF** (1 major + 2 minor)
B, 11 y	Polyarthralgia, ESR 45, PR 0.20 s; echo normal; ASO high	Polyarthralgia (1)	ESR, PR (2)	Yes	**ARF** (high-risk rules), if other causes excluded
C, 10 y	Jerky movements, emotional lability, ASO normal, echo shows mild MR	Chorea (alone is enough) + subclinical carditis	—	Not required	**ARF (chorea) with carditis**
D, 14 y, known RHD	Fever 38.3, monoarthralgia, ESR 50; ASO high	0	3	Yes	**Recurrent ARF** (3 minor allowed for recurrence)

Differential diagnosis

Condition	Distinguishing features	Key investigation
Septic arthritis	Single hot joint, very unwell, cannot bear weight, not migratory	Joint aspiration, blood culture
Sickle cell crisis	Known SCD, bone pain, dactylitis, anaemia, jaundice	Hb electrophoresis, FBC
Post-streptococcal reactive arthritis	Arthritis within 10 days of GAS, non-migratory, poor response to NSAIDs, no carditis	Clinical; echo normal (some still give prophylaxis)
Juvenile idiopathic arthritis	Arthritis >6 weeks, morning stiffness, uveitis, salmon rash (systemic JIA)	Clinical; ANA, RF
Leukaemia	Bone pain, pallor, bruising, organomegaly	FBC/film
Infective endocarditis	Known valve disease, fever, embolic phenomena	Blood cultures, echo
Viral myocarditis / dilated cardiomyopathy	Heart failure without valvulitis, poor LV function, no joint signs	Echo
Congenital heart disease (VSD, congenital MR)	Murmur from infancy, failure to thrive from early life	Echo
Lyme, SLE, HSP	Rare here; rash, renal involvement	Specific tests
Other chorea: drugs, SLE, Huntington, Wilson	Family history, drug history	Clinical, specific tests

Investigations

Test	What you are looking for	Why
ESR/CRP	Inflammation	Minor criterion; monitor activity
ASO / anti-DNase B / throat culture	Recent GAS	Required evidence (except chorea/indolent carditis)
ECG	Prolonged PR, AF, block	Minor criterion; rhythm
Echo	Carditis, valve damage	Major criterion; staging and duration of prophylaxis
CXR	Cardiomegaly, congestion	Heart failure
FBC	Anaemia, leukaemia	Alternative diagnoses; treat anaemia
Blood cultures	Endocarditis	Fever in known RHD

Management

A. Acute rheumatic fever: step by step

1. **Admit** (UCG: HC IV and above; WHO: admit) for diagnosis, echo and education. Bed rest while inflammation and failure are active; then **gradual mobilisation** as CRP falls and symptoms settle (strict prolonged bed rest is no longer advised).
2. **Eradicate GAS now**, even if the throat swab is negative, with **one dose of BPG**, which also counts as the **first prophylactic dose**.
3. **Treat arthritis/fever** with aspirin or naproxen (do not start before the diagnosis is clear, if possible, because they hide the migratory pattern; paracetamol is fine while you observe).
4. **Echo** for every child.
5. **Treat heart failure** if present.
6. **Treat chorea** if disabling.
7. **Start secondary prophylaxis**, give the family a **BPG card**, book the next injection, and register the child in the **RHD register/clinic**.
8. **Educate**: what ARF is, why injections every 4 weeks matter, signs of heart failure, dental care, contraception and pregnancy planning for girls.

B. Drugs for ARF

Purpose	Drug	Dose	Route	Frequency	Duration	Notes
Eradicate GAS	**Benzathine penicillin G**	**0.6 MU (600,000 IU) if <30 kg; 1.2 MU if ≥30 kg** (UCG 2023; AHA cut-off 27 kg)	**Deep IM** (gluteal or lateral thigh)	Single dose	—	Preferred. Counts as first prophylactic dose
	or **Penicillin V**	**Child: 250 mg** (UCG lists 125 mg/dose for young children; 500 mg if ≥30 kg)	Oral	6–12-hourly (UCG 6-hourly)	**10 days**	Adherence for 10 days is poor

Purpose	Drug	Dose	Route	Frequency	Duration	Notes
	or **Amoxicillin**	**50 mg/kg once daily (max 1 g)**	Oral	Once daily	**10 days**	Good alternative (AHA)
	Penicillin allergy: **erythromycin**	10–12.5 mg/kg (max 500 mg)	Oral	6-hourly	10 days	Or azithromycin 12 mg/kg once daily (max 500 mg) for 5 days
Arthritis, fever	**Aspirin**	**UCG 2023: 80–100 mg/kg/day in 3 doses** (WHO Pocket Book: 20 mg/kg 6-hourly for 1–2 weeks, then 15 mg/kg 6-hourly for 3–6 more weeks)	Oral, with food	6–8-hourly	Until joints settle and CRP falls (usually **2–6 weeks**, UCG 4–8 weeks), then taper	**Max about 4 g/day.** Toxicity: tinnitus, vomiting, hyperventilation; check salicylate level if available. Reye syndrome risk with influenza/varicella. UCG adds an antacid
	or **Naproxen**	**10–20 mg/kg/day in 2 doses (max 1 g/day)**	Oral	12-hourly	As above	Preferred in some guidelines (Australia 2020): fewer side effects, twice daily
	or **Ibuprofen**	10 mg/kg (max 400–600 mg)	Oral	8-hourly	As above	Alternative
Severe carditis / heart failure (optional, specialist)	**Prednisolone**	**1–2 mg/kg/day (max 60–80 mg)** (WHO Pocket Book: 1 mg/kg/day for 1 week in severe heart failure)	Oral	Once daily	1–3 weeks, then **taper by ~25% per week**	Weak evidence; WHO 2024 makes **no recommendation** for or against anti-inflammatories to prevent RHD. Overlap with aspirin during taper to avoid rebound
Chorea (if disabling)	**Sodium valproate**	**10–20 mg/kg/day** (UCG), up to 30 mg/kg/day in 2–3 doses	Oral	8–12-hourly	Until controlled for several weeks, then taper	Or **carbamazepine** 7–20 mg/kg/day in 2–3 doses. Haloperidol is an alternative (extrapyramidal effects). Calm environment

DRUG DOSING

BPG technique: use the **correct dose for weight**, warm the vial to room temperature, inject **slowly (over 2–3 minutes)** deep IM, and consider **lidocaine as diluent** (WHO 2024 conditional recommendation) to reduce pain. Applying pressure or ice beforehand and distraction help. Observe the child for **15–30 minutes** afterwards. Children with **severe valve disease** (severe MS/AR, pulmonary hypertension, poor LV function) can collapse after a **vasovagal** reaction to the painful injection, which is often mistaken for anaphylaxis: give it **lying down**, with an experienced clinician nearby, or discuss oral prophylaxis with cardiology.

C. Heart failure from carditis or chronic RHD

General measures: sit up (semi-recumbent), oxygen if SpO_2 <90% or distressed (WHO: aim >90%), restrict IV fluids, salt restriction, treat fever and anaemia (transfuse packed cells slowly with furosemide if Hb <7–8 g/dL; WHO Pocket Book says transfuse if Hb <8 g/dL in rheumatic heart failure), treat infection, weigh daily.

Drug	Dose	Route	Frequency	Max	Notes
Furosemide (acute)	**1 mg/kg**; if no response in 2 h, **2 mg/kg** (WHO Pocket Book). UCG acute: 0.5–1.5 mg/kg/dose	IV (faster) or oral	6–12-hourly as needed, then 1–2 mg/kg once daily maintenance	**6 mg/kg/day** (UCG)	Monitor K^+, urine output, weight
Spironolactone	**1–3 mg/kg/day** (UCG 1.5–3 mg/kg/day)	Oral	Once or twice daily	100 mg/day	K^+-sparing; check K^+ with ACE inhibitor
Potassium chloride (if furosemide alone >5 days, or with digoxin)	**3–5 mmol/kg/day** (WHO Pocket Book)	Oral	Divided	—	Not needed if on spironolactone or ACE inhibitor
Enalapril	Start **0.05–0.1 mg/kg/day**, titrate over 2 weeks to **0.5 mg/kg/day** (UCG range 0.1–1 mg/kg/day)	Oral	Once or twice daily	**20 mg/day** in children (40 mg adults)	For MR/AR with LV dilatation. **Avoid in severe AS/MS-dominant disease with low BP.** Check creatinine and K^+. Teratogenic
or **Captopril**	**0.1–0.3 mg/kg/dose** (start low), titrate	Oral	8-hourly	6 mg/kg/day (max 150 mg/day)	Test dose with BP monitoring
Digoxin (AF with fast rate, or poor LV function)	Loading **15 µg/kg** oral, then **5 µg/kg** starting 6 h later (WHO Pocket Book); maintenance **5 µg/kg**	Oral	Maintenance **12-hourly**	**250 µg per dose**	Toxic with hypokalaemia; check pulse before each dose (hold if bradycardic); no loading if on digoxin already
Carvedilol (stable chronic failure, specialist)	0.05 mg/kg titrated to 0.35 mg/kg (UCG)	Oral	12-hourly	25 mg/dose	Only when euvolaemic and stable
Warfarin (AF, LA thrombus, mechanical valve, prior embolism)	Specialist dosing	Oral	Daily	INR 2–3 (AF); 2.5–3.5 (mechanical mitral)	INR monitoring is difficult outside referral centres; drug interactions; teratogenic in the first trimester

UGANDA CONTEXT

At HC IV/general hospital level you can start **furosemide, spironolactone, an ACE inhibitor (enalapril or captopril), BPG and digoxin** (UCG levels of care HC4/H). **Echo, warfarin monitoring and referral for intervention** happen at RRH cardiac clinics and the **Uganda Heart Institute**. Refer any child with severe valve disease, heart failure despite treatment, AF, or suspected endocarditis.

D. Secondary prophylaxis (the most important step)

Regimen	Dose	Frequency	Notes
Benzathine penicillin G (first line; WHO 2024 strong recommendation)	**0.6 MU if <30 kg; 1.2 MU if ≥30 kg** (UCG 2023)	**Every 4 weeks** (WHO, UCG, AHA); **every 3 weeks** if breakthrough ARF on 4-weekly dosing or very high-risk (AHA)	About **10 times more effective** than oral penicillin for preventing recurrence (Cochrane). Keep a card; trace defaulters

Regimen	Dose	Frequency	Notes
Penicillin V (if injections not possible)	**250 mg** (UCG child 125–250 mg; adolescents 500 mg)	**12-hourly**, daily	Less effective; adherence is hard. WHO 2024: acceptable alternative
Erythromycin (true penicillin allergy)	**250 mg** (UCG child: 10 mg/kg)	**12-hourly**	For anaphylaxis/severe allergy history. WHO 2024: for mild allergy history, consider an **oral penicillin test dose** before BPG; **routine skin testing is not recommended** in those without allergy history

Duration of prophylaxis

Category	UCG 2023	AHA/ACC 2020
ARF **without carditis**	5 years or until age **18–21**	5 years or until age 21, **whichever is longer**
ARF **with carditis but no residual valve disease**	10 years or until age **25**	10 years or until age 21, whichever is longer
ARF **with carditis and residual valve disease** (RHD)	Until age **40–45 or for life**	10 years or until age 40, whichever is longer (sometimes lifelong)
Latent / screen-detected RHD (children)	—	WHO 2024: give prophylaxis (GOAL: 4-weekly BPG; re-echo at 2 years and decide)
After valve surgery	For life (UCG: RHD prophylaxis for life)	Continue per category above

EXAM FAVOURITE

GOAL trial (Beaton et al., *NEJM* 2022; Ugandan children aged 5–17 with **latent RHD**): BPG every 4 weeks for 2 years reduced echo progression from **8.2% to 0.8%.** It is the evidence behind WHO 2024's recommendation to give prophylaxis to children with screen-detected RHD.

E. Primary prevention: the sore throat

- **WHO 2024**: treat **confirmed GAS pharyngitis** with antibiotics (strong); in **moderate/high-risk settings without testing**, treat **clinically suspected GAS pharyngitis in children and adolescents** (strong); **penicillin (IM or oral)** is first line.
- In Uganda: **BPG single dose** (0.6 MU <30 kg, 1.2 MU ≥30 kg) or **amoxicillin 50 mg/kg once daily (max 1 g) for 10 days** or penicillin V for 10 days.
- Treat household contacts with sore throats; reduce crowding; health education in schools.

F. Infective endocarditis prevention and dental care

- **Good dental hygiene** and regular dental review matter more than antibiotic prophylaxis.
- **AHA**: antibiotic prophylaxis before dental procedures that manipulate the gums is now advised only for the **highest-risk** patients: **prosthetic valves or prosthetic material for valve repair, previous endocarditis**, certain congenital heart disease and transplant valvulopathy. Uncomplicated RHD alone is no longer an indication (AHA), though some programmes in endemic countries still give it.
- Regimen: **amoxicillin 50 mg/kg (max 2 g) orally 30–60 minutes before** the procedure (clindamycin 20 mg/kg, max 600 mg, or azithromycin 15 mg/kg, max 500 mg, if allergic). **Note**: a child on long-term penicillin prophylaxis may carry penicillin-resistant mouth streptococci, so the AHA suggests a non-penicillin agent (clindamycin or azithromycin) for these children (specialist advice).

G. Interventions for chronic RHD (UHI)

Lesion	Intervention
Severe MS with pliable valve	**Percutaneous balloon mitral valvotomy**
Severe MR/AR with symptoms or LV dilatation/dysfunction	**Valve repair** (preferred in children) or **replacement** (mechanical needs lifelong warfarin; bioprosthetic valves degenerate fast in the young)
Any surgery	Continue BPG prophylaxis; anticoagulation clinic; family planning (warfarin is teratogenic)

Complications

Acute	Chronic
Heart failure, pulmonary oedema	**Chronic RHD**: MR, MS, AR, AS, mixed lesions
Pericardial effusion (tamponade rare)	Atrial fibrillation, stroke and systemic emboli
Heart block	Pulmonary hypertension, right heart failure
Death from fulminant carditis (rare)	**Infective endocarditis**
Disabling chorea, emotional disturbance (usually resolves in weeks to months)	Recurrence of ARF with further valve damage
	High maternal mortality in pregnancy; growth failure; school absence; premature death

Ward-round & exam pearls

Model case presentation

"**Akello is an 11-year-old girl weighing 27 kg** (weight-for-age Z –1.2, height-for-age Z –0.6, BMI-for-age Z –1.3) from a boarding primary school in Lira. She is **immunised per UNEPI**, eats 3 meals a day at school, and is **HIV-negative**. She presents with **2 weeks of joint pains that moved from the right knee to the left ankle and then the right elbow**, with fever, after a sore throat about 3 weeks ago for which she got no treatment. She has had **exertional breathlessness for 4 days**. On examination: temperature 38.3 °C, HR 128 (also 120 while asleep), RR 30, BP 98/56, SpO_2 96%. Her left ankle is swollen, warm and very tender. There is an **apical grade 3/6 pansystolic murmur radiating to the axilla** and a soft apical mid-diastolic murmur, no gallop, and the liver is 2 cm. **ESR 85 mm/h, CRP 96 mg/L, ASO 800 IU/mL, PR interval 0.18 s.** Echo shows **moderate MR with thickened anterior mitral leaflet tips**. Using the 2015 high-risk Jones criteria she has **2 major criteria** (polyarthritis, carditis) and **2 minor criteria** (fever, raised ESR/CRP; the long PR does not count because carditis is already a major criterion), with GAS evidence (ASO): **initial ARF with moderate carditis**. Plan: **BPG 0.6 MU IM today** (she is under 30 kg), **aspirin 80–100 mg/kg/day in 3–4 doses** (or naproxen), monitor for heart failure, register for **4-weekly BPG for at least 10 years or until age 25** (UCG; lifelong if residual disease), and counsel the family."

Viva questions (with answers)

1. **What changed in the 2015 Jones criteria?** Stratification into low-risk and moderate/high-risk populations; **subclinical carditis on echo** counts as major; in high-risk populations **monoarthritis and polyarthralgia** count as major, with lower fever (≥38.0) and ESR (≥30) thresholds for minor criteria; **3 minor** can diagnose recurrence.

2. **Why is ASO often normal in chorea?** Chorea appears 1–6 months after GAS infection, when ASO has fallen; chorea alone is diagnostic.
3. **Which valve is most often affected and how does it present in children?** The mitral valve, usually as **regurgitation** (apical pansystolic murmur to the axilla).
4. **What is the most effective intervention to prevent RHD progression?** Regular **BPG every 4 weeks** (secondary prophylaxis).
5. **How long do you give prophylaxis to a child with carditis that resolved?** UCG: 10 years or until age 25; AHA: 10 years or until 21, whichever is longer.
6. **What is a Carey Coombs murmur?** A soft apical mid-diastolic murmur from high flow across an inflamed mitral valve in acute carditis (not true stenosis).

OSCE tips

- In a chorea station, demonstrate **milkmaid's grip, pronator sign, darting tongue** and ask the child to write or draw a spiral.
- In a counselling station on BPG: explain why, how often, how long, what to do if a dose is missed, pain relief, and that **the injection protects the heart**.

Mnemonics & memory aids

- **Major criteria "J♥NES"**: **J**oints, ♥ (carditis), **N**odules, **E**rythema marginatum, **S**ydenham chorea.
- **Minor criteria "CAFE PAL"** (classic): **C**RP, **A**rthralgia, **F**ever, **E**SR, **P**rolonged PR, **A**SO (supporting evidence, not a minor criterion), **L**eucocytosis (WHO Pocket Book lists it). Know that the 2015 version uses fever, arthralgia, ESR/CRP and PR.
- **"Licks the joints, bites the heart"** (Lasègue).
- **Prophylaxis duration "5–10–40"**: no carditis 5 years; carditis without residual disease 10 years; residual valve disease until about 40 (or for life).
- **BPG "30 kg divides 0.6 and 1.2"** (UCG).

High-yield summary

Numbers & doses to know

Item	Value
Latent period after GAS pharyngitis	2–4 weeks (chorea 1–6 months)
Peak age	5–15 years
High-risk minor thresholds	Fever ≥38.0 °C; ESR ≥30 mm/h; CRP ≥3.0 mg/dL (30 mg/L)
Low-risk minor thresholds	Fever ≥38.5 °C; ESR ≥60 mm/h
PR upper limit (3–12 y / 12–16 y / ≥17 y)	0.16 / 0.18 / 0.20 s
BPG dose (UCG)	0.6 MU <30 kg; 1.2 MU ≥30 kg, deep IM
BPG frequency	Every 4 weeks (3 weeks if breakthrough)
Amoxicillin for GAS sore throat	50 mg/kg once daily (max 1 g) × 10 days
Aspirin (UCG)	80–100 mg/kg/day in 3 doses
Naproxen	10–20 mg/kg/day in 2 doses (max 1 g/day)
Prednisolone (severe carditis)	1–2 mg/kg/day (max 60–80 mg), taper

Item	Value
Furosemide (WHO)	1 mg/kg, then 2 mg/kg if no response
Enalapril	Start 0.05–0.1 mg/kg/day; up to 0.5 mg/kg/day (max 20 mg)
Digoxin (WHO)	Load 15 µg/kg; maintain 5 µg/kg 12-hourly (max 250 µg/dose)
GOAL trial progression	0.8% (BPG) vs 8.2% (none) at 2 years
WHO global burden	~55 million with RHD; ~360,000 deaths/year

Classic exam traps

- **Trap**: Requiring polyarthritis for a Ugandan child. **Right**: in high-risk populations monoarthritis and polyarthralgia can count as major.
- **Trap**: "ASO is normal, so it can't be ARF" in a child with chorea. **Right**: chorea alone is diagnostic.
- **Trap**: Starting aspirin before the diagnosis is clear. **Right**: NSAIDs mask migratory arthritis; use paracetamol while observing (if the child is stable).
- **Trap**: Treating ARF and forgetting secondary prophylaxis. **Right**: give BPG at diagnosis and book the next one in 4 weeks.
- **Trap**: "Oral penicillin is as good as BPG." **Right**: BPG is about 10 times more effective; oral is a fallback.
- **Trap**: Labelling a fainting child after BPG as anaphylaxis. **Right**: in severe valve disease it is often vasovagal; give BPG lying down and observe.
- **Trap**: Giving ACE inhibitors to a hypotensive child with severe mitral stenosis. **Right**: MS needs rate control and diuretics, then mechanical relief.

Questions seniors ask

1. **What is the evidence of streptococcal infection you accept?** Raised/rising ASO or anti-DNase B, positive throat culture or rapid antigen test, recent scarlet fever.
2. **Why do we echo every child with ARF?** To detect **subclinical carditis**, which is a major criterion and lengthens prophylaxis.
3. **How does molecular mimicry explain ARF?** Antibodies to streptococcal M protein and N-acetylglucosamine cross-react with cardiac myosin, valve laminin, synovium and basal ganglia.
4. **What did WHO's 2024 guideline add?** Endorses Jones criteria; handheld echo for diagnosis; echo screening of children 5–19 in high-prevalence settings; prophylaxis for screen-detected RHD; BPG first line; oral penicillin acceptable alternative; lidocaine diluent; adherence support; no recommendation on anti-inflammatories.
5. **When do you refer to the Uganda Heart Institute?** Moderate–severe valve disease, heart failure not controlled, AF or embolism, suspected endocarditis, need for valvotomy or surgery, pregnancy with RHD.
6. **How would you improve adherence to BPG?** Pain reduction (lidocaine, slow injection), reminder systems, cards, community health worker tracing, giving injections close to home, and education of the child and school.

SELF-TEST

Flashcards

26 cards. Cover the right-hand column and test yourself.

#	Question	Answer
1	What is acute rheumatic fever?	An **autoimmune** inflammatory illness about **2–4 weeks after group A streptococcal pharyngitis** (± skin infection), affecting joints, heart, brain, skin and subcutaneous tissue; mainly ages **5–15**.
2	Jones major criteria (2015)?	**J♥NES**: Joints (arthritis), **Carditis** (clinical or subclinical), **Nodules** (subcutaneous), **Erythema marginatum**, **Sydenham chorea**.
3	Minor criteria in moderate/high-risk populations (e.g. Uganda)?	**Monoarthralgia**, **fever ≥38.0 °C**, **ESR ≥30 mm/h and/or CRP ≥3.0 mg/dL (30 mg/L)**, **prolonged PR** (age-adjusted; not if carditis is a major criterion).
4	What joint findings count as MAJOR in high-risk populations?	**Monoarthritis or polyarthritis**, and/or **polyarthralgia** (if other causes excluded). In low-risk populations only polyarthritis counts.
5	Diagnostic rule for initial vs recurrent ARF?	GAS evidence + **initial**: 2 major, or 1 major + 2 minor. **Recurrent**: 2 major, 1 major + 2 minor, or **3 minor**.
6	Which manifestations diagnose ARF on their own?	**Sydenham chorea** and **indolent (insidious) carditis**: no other criteria or streptococcal evidence needed.
7	Acceptable evidence of preceding GAS infection?	Raised or rising **ASO** or **anti-DNase B**, positive **throat culture**, positive **rapid GAS antigen test**, recent **scarlet fever**.
8	Age-adjusted upper limit of the PR interval?	**3–12 y: 0.16 s; 12–16 y: 0.18 s; ≥17 y: 0.20 s.**
9	WHF Doppler criteria for pathological mitral regurgitation?	Seen in **2 views**, jet **≥2 cm** in at least one view, peak velocity **≥3 m/s**, **pansystolic** in at least one envelope.
10	Mechanism of ARF?	**Molecular mimicry**: antibodies/T cells against streptococcal **M protein** and **N-acetylglucosamine** cross-react with **cardiac myosin, valve laminin, synovium and basal ganglia**.
11	Which valves does RHD affect, in order?	**Mitral** (most; regurgitation in children, stenosis later) > **mitral + aortic** > aortic alone > tricuspid (rarely isolated) > pulmonary (almost never).
12	What is a Carey Coombs murmur?	A soft **apical mid-diastolic** murmur in **acute mitral valvulitis** (high flow over an inflamed valve), without an opening snap.
13	Sydenham chorea: signs?	Involuntary jerky movements, **milkmaid's grip**, **darting tongue**, **pronator sign**, spooning of hands, hypotonia, **emotional lability**; absent in sleep. Appears 1–6 months after GAS infection.
14	BPG dose for a child (UCG 2023)?	**0.6 MU (600,000 IU) if <30 kg; 1.2 MU if ≥30 kg**, deep IM. (AHA uses a 27 kg cut-off.)
15	How often is BPG given for secondary prophylaxis?	**Every 4 weeks** (UCG, WHO 2024, AHA); **every 3 weeks** if breakthrough ARF or very high risk (AHA).
16	Duration of prophylaxis (UCG 2023)?	No carditis: **5 years or until age 18–21**. Carditis, no residual disease: **10 years or until 25**. Residual valve disease: **until age 40–45 or for life**.
17	Duration of prophylaxis (AHA/ACC 2020)?	No carditis: 5 y or age 21; carditis without residual disease: 10 y or age 21; carditis with residual disease: 10 y or age 40 — **whichever is longer**.
18	What did the GOAL trial show?	In Ugandan children 5–17 years with **latent RHD**, BPG every 4 weeks for 2 years reduced echo progression from **8.2% to 0.8%** (NEJM 2022).

#	Question	Answer
19	Aspirin dose in ARF arthritis (UCG 2023 and WHO Pocket Book)?	UCG: **80–100 mg/kg/day in 3 doses**. WHO: **20 mg/kg 6-hourly** for 1–2 weeks, then 15 mg/kg 6-hourly for 3–6 weeks. Max about 4 g/day; watch for tinnitus and hyperventilation.
20	Alternative NSAID to aspirin in ARF and its dose?	**Naproxen 10–20 mg/kg/day in 2 divided doses (max 1 g/day)**; ibuprofen also used.
21	Arthritis not improving within 72 hours of NSAIDs: what does it suggest?	Reconsider the diagnosis: septic arthritis, post-streptococcal reactive arthritis, JIA, SCD, leukaemia.
22	Heart failure drugs in rheumatic carditis (with doses)?	**Furosemide 1 mg/kg** (then 2 mg/kg if no response; WHO), **spironolactone 1–3 mg/kg/day**, **enalapril** start 0.05–0.1 mg/kg/day titrating to ~0.5 mg/kg/day (max 20 mg), **digoxin** load 15 µg/kg then 5 µg/kg 12-hourly. Prednisolone 1–2 mg/kg/day for severe carditis (weak evidence).
23	WHO 2024 guideline: first-line drug for GAS pharyngitis and for RF/RHD prophylaxis?	**Penicillin** (IM or oral) for GAS pharyngitis; **IM BPG** for secondary prophylaxis, with **oral penicillin** as an acceptable alternative.
24	WHO 2024 on echo screening?	In moderate/high RHD prevalence settings, **echo screening (standard or handheld) of children and adolescents 5–19 years** may be implemented (strong rec); **handheld echo** can diagnose RF carditis/RHD where standard echo is unavailable.
25	Why give BPG lying down to a child with severe valve disease?	Painful injection can trigger a **vasovagal** reaction; in severe MS/AR or pulmonary hypertension this can cause collapse, often misread as anaphylaxis. Observe for 15–30 minutes.
26	Infective endocarditis prophylaxis for dental procedures (AHA) — who and what?	Only highest-risk: **prosthetic valves/material, previous endocarditis**, certain CHD. **Amoxicillin 50 mg/kg (max 2 g)** 30–60 min before; clindamycin/azithromycin if on long-term penicillin or allergic. Dental hygiene for all.

SELF-TEST

Single best answer questions

15 questions. The answers and explanations follow at the end.

1. A 10-year-old boy (26 kg) from Gulu has a hot, swollen, very painful right knee for 5 days, fever 38.2 °C and CRP 64 mg/L. ASO is 600 IU/mL. Echo is normal. The knee aspirate is sterile with no organisms. Which statement BEST applies the 2015 Jones criteria?

A ARF cannot be diagnosed because polyarthritis is required

B ARF cannot be diagnosed without carditis

C He has post-streptococcal reactive arthritis by definition

D He meets criteria for initial ARF: monoarthritis is a major criterion in high-risk populations, plus 2 minor (fever ≥38.0 °C and raised CRP), with GAS evidence

E He needs 3 minor criteria because this is a first episode

2. A 12-year-old girl (34 kg) is diagnosed with initial ARF with polyarthritis and moderate mitral regurgitation on echo. What is the MOST important prescription to prevent future valve damage?

A Aspirin 100 mg/kg/day for 6 months

B Benzathine penicillin G 1.2 MU IM now and every 4 weeks, for at least 10 years or until age 25 (UCG), lifelong if residual valve disease

C Penicillin V 250 mg twice daily for 10 days only

D Prednisolone 2 mg/kg/day for 3 months

E Amoxicillin 50 mg/kg before dental procedures only

3. A 9-year-old girl is brought because her handwriting has deteriorated, she drops cups, and she cries easily. On exam she has jerky movements of the face and hands, milkmaid's grip and a positive pronator sign. ASO is normal; ESR is 18 mm/h. Echo shows mild pathological MR. What is the diagnosis?

A Normal behaviour for age

B Tic disorder

C Wilson disease

D Not ARF, because ASO is normal

E Acute rheumatic fever (Sydenham chorea, with subclinical carditis)

4. A 13-year-old boy (30 kg) with known RHD (severe MR and AR) presents with orthopnoea, RR 44, HR 136, gallop, liver 5 cm and bilateral basal crackles. SpO_2 88%. BP 104/40. What is the MOST appropriate immediate management?

A Sit him up, give oxygen, IV furosemide 1 mg/kg (30 mg), restrict fluids and plan an ACE inhibitor once stable

B Normal saline 20 mL/kg bolus for the low diastolic BP

C IV propranolol for the tachycardia

D Aspirin 100 mg/kg/day

E Benzathine penicillin only and discharge

5. A 7-year-old (21 kg) with sore throat, fever 38.6 °C, tender anterior cervical nodes, tonsillar exudate and no cough presents to a HC III in Lira. No rapid strep test is available. According to WHO 2024, what is the BEST management?

A No antibiotics, because the cause cannot be confirmed
B Ciprofloxacin for 5 days
C Treat with penicillin: BPG 0.6 MU IM once (or amoxicillin 50 mg/kg once daily for 10 days)
D Wait 2 weeks and treat only if joint pains develop
E Azithromycin as first line for everyone

6. A 14-year-old girl on BPG for RHD has a documented history of an itchy rash (no swelling or breathing difficulty) after amoxicillin at age 5. She has no other allergy history. What does WHO 2024 suggest? Advanced

A Stop all prophylaxis
B Consider an oral penicillin test dose before IM BPG, as there is no history of anaphylaxis, angioedema or SJS/TEN
C Routine penicillin skin testing for every child on BPG
D Switch to ciprofloxacin prophylaxis
E Give BPG with prophylactic adrenaline every month

7. A 15-year-old with severe rheumatic mitral stenosis and pulmonary hypertension becomes pale, sweaty and hypotensive with HR 48/min two minutes after a BPG injection. There is no rash, wheeze or stridor. What is the MOST likely explanation and action? Advanced

A Anaphylaxis; give adrenaline and never give penicillin again
B Acute haemolytic reaction; send blood to the lab
C Infective endocarditis; start antibiotics
D Vasovagal reaction; lie flat with legs raised, monitor, and give subsequent injections lying down with a slower injection and pain relief
E Pulmonary embolism; start heparin

8. An 11-year-old with ARF and polyarthritis is started on aspirin 25 mg/kg 6-hourly. On day 3 she has nausea, ringing in the ears and is breathing fast with a clear chest. What is the MOST likely cause?

A Salicylate toxicity
B Worsening carditis
C Reye syndrome
D Aspiration pneumonia
E Sydenham chorea

9. School echo screening in Lira finds an asymptomatic 9-year-old (24 kg) with mild pathological mitral regurgitation meeting minimum criteria for RHD (latent RHD). No history of ARF. What is the MOST appropriate management?

A Reassure and discharge; no treatment for asymptomatic children
B Start enalapril and furosemide
C Refer for mitral valve replacement
D Repeat ASO every month and treat only if it rises
E Start BPG 0.6 MU IM every 4 weeks, enrol in the RHD clinic and repeat echo (e.g. at 2 years)

10. A 12-year-old (32 kg) had ARF with carditis 2 years ago; echo now shows no residual valve disease. According to UCG 2023, how long should BPG prophylaxis continue?

A Until 5 years after the attack only
B Until age 21 only
C For 10 years or until age 25
D For life regardless of findings
E It can stop now since the echo is normal

11. A 14-year-old with known RHD on monthly BPG has had fever for 3 weeks, weight loss, a new diastolic murmur, splinter haemorrhages, splenomegaly and microscopic haematuria. What is the MOST appropriate next step?

A Take three sets of blood cultures before starting antibiotics, and arrange echocardiography
B Start aspirin for recurrent ARF
C Give a single dose of BPG and send home
D Start antimalarials and review in a week
E Start prednisolone for carditis

12. A 10-year-old with ARF (polyarthritis, mild carditis) is on aspirin 80 mg/kg/day. After 72 hours there is no improvement in her joint pain or swelling. What is the BEST next step? Advanced

A Double the aspirin dose
B Add prednisolone immediately
C Continue the same dose for 2 more weeks before reassessing
D Reconsider the diagnosis (e.g. septic arthritis, JIA, sickle cell disease, leukaemia, post-streptococcal reactive arthritis)
E Stop BPG

13. Which auscultatory finding is MOST typical of acute rheumatic carditis in a child?

A A continuous machinery murmur below the left clavicle
B A fixed widely split second heart sound
C An apical pansystolic murmur radiating to the axilla
D An ejection systolic murmur at the upper left sternal edge with an ejection click
E A venous hum that disappears lying flat

14. A 16-year-old girl with RHD and a mechanical mitral valve on warfarin asks about contraception and future pregnancy. Which advice is MOST appropriate? Advanced

A She can stop warfarin when she becomes pregnant
B Pregnancy is low risk with a mechanical valve
C Oestrogen-containing pills are the best choice
D Warfarin is safe throughout pregnancy
E Use effective contraception (e.g. a copper IUD, implant or progestogen-only method), plan any pregnancy with the cardiologist and obstetrician, because warfarin is teratogenic in early pregnancy and mechanical valves carry high thrombosis risk

15. Which statement about the 2015 revised Jones criteria is CORRECT?

A Subclinical carditis on echo does not count as a criterion
B In moderate/high-risk populations, an ESR ≥30 mm/h counts as a minor criterion
C A prolonged PR interval counts as a minor criterion even when carditis is a major criterion
D Evidence of GAS infection is required for chorea
E Fever must be ≥39 °C to count as a minor criterion

Answers & explanations

1. D · He meets criteria for initial ARF: monoarthritis is a major criterion in high-risk populations, plus 2 minor (fever ≥38.0 °C and raised CRP), with GAS evidence

D is correct: Uganda is a moderate/high-risk setting, so **monoarthritis** counts as major; fever ≥38.0 °C and CRP ≥30 mg/L are minor; ASO provides GAS evidence: 1 major + 2 minor. A is wrong: polyarthritis is required only in low-risk populations. B is wrong: carditis is not mandatory. C is wrong: reactive arthritis is a diagnosis of exclusion and Jones criteria are met here. E is wrong: 3 minor criteria apply to recurrences, not initial episodes.

2. B · Benzathine penicillin G 1.2 MU IM now and every 4 weeks, for at least 10 years or until age 25 (UCG), lifelong if residual valve disease

B is correct: **secondary prophylaxis with BPG** every 4 weeks prevents recurrences that worsen RHD; at ≥30 kg the dose is 1.2 MU, and with carditis the duration is long (residual disease: until 40–45 or life). A is wrong: aspirin treats arthritis but does not prevent RHD, and 6 months is excessive. C is wrong: 10 days of penicillin only eradicates current GAS. D is wrong: steroids have weak evidence and no effect on recurrence. E is wrong: endocarditis prophylaxis does not prevent ARF recurrence.

3. E · Acute rheumatic fever (Sydenham chorea, with subclinical carditis)

E is correct: **Sydenham chorea is diagnostic of ARF on its own**, and appears 1–6 months after GAS when ASO may have normalised; the echo also shows subclinical carditis, which lengthens prophylaxis. A is wrong: these are neurological signs. B is wrong: tics are stereotyped and suppressible, without milkmaid's grip or hypotonia. C is wrong: Wilson disease is rare at this age and causes liver disease and Kayser–Fleischer rings. D is wrong: GAS evidence is not required for chorea.

4. A · Sit him up, give oxygen, IV furosemide 1 mg/kg (30 mg), restrict fluids and plan an ACE inhibitor once stable

A is correct: he has acute **heart failure** from regurgitant lesions; sit up, oxygen to SpO_2 >90%, IV furosemide (WHO 1 mg/kg, then 2 mg/kg if no response), restrict fluids, then afterload reduction with an ACE inhibitor for MR/AR. B is wrong: the low diastolic BP is from aortic regurgitation; fluid would worsen pulmonary oedema. C is wrong: the tachycardia is compensatory; IV beta-blockade can cause collapse. D is wrong: there is no evidence of active ARF arthritis here. E is wrong: BPG is essential but does not treat acute failure, and he needs admission.

5. C · Treat with penicillin: BPG 0.6 MU IM once (or amoxicillin 50 mg/kg once daily for 10 days)

C is correct: WHO 2024 recommends that in **moderate/high-risk settings without testing**, children with clinically suspected GAS pharyngitis receive antibiotics, with **penicillin (IM or oral) first line**; under 30 kg the BPG dose is 0.6 MU. A is wrong: that misses primary prevention. B is wrong: ciprofloxacin has poor GAS activity. D is wrong: treating after ARF starts does not prevent it. E is wrong: macrolides are for penicillin allergy; resistance is a concern.

6. B · Consider an oral penicillin test dose before IM BPG, as there is no history of anaphylaxis, angioedema or SJS/TEN

B is correct: WHO 2024 conditionally recommends an **oral penicillin test dose** before BPG in patients with a history of **mild** penicillin allergy (no anaphylaxis, angioedema or SJS/TEN). A is wrong: stopping prophylaxis exposes her to recurrence. C is wrong: WHO advises against allergy testing in patients **without** an allergy history, and testing is not routine for all. D is wrong: ciprofloxacin is not an anti-streptococcal prophylactic. E is wrong: prophylactic adrenaline is not indicated.

7. D · Vasovagal reaction; lie flat with legs raised, monitor, and give subsequent injections lying down with a slower injection and pain relief

D is correct: **bradycardia with hypotension and pallor without rash, wheeze or stridor** after a painful injection is a **vasovagal** reaction, which can be dangerous in severe valve disease; manage supine with legs raised, and prevent next time (lying down, lidocaine diluent, slow injection). A is wrong: anaphylaxis typically causes tachycardia with skin or airway signs (though if in doubt, adrenaline is not withheld). B is wrong: no blood was transfused. C is wrong: endocarditis is subacute, not a 2-minute event. E is wrong: nothing suggests PE and heparin is not indicated.

8. A · Salicylate toxicity

A is correct: **tinnitus, nausea and hyperventilation** (respiratory stimulation) are classic **salicylate toxicity**; reduce or stop the dose, check a salicylate level if possible, or switch to naproxen. B is wrong: carditis would cause tachycardia, a gallop and crackles, not tinnitus. C is wrong: Reye syndrome causes encephalopathy and liver failure, usually after a viral illness. D is wrong: the chest is clear. E is wrong: chorea causes movements, not tinnitus or tachypnoea.

9. E · Start BPG 0.6 MU IM every 4 weeks, enrol in the RHD clinic and repeat echo (e.g. at 2 years)

E is correct: WHO 2024 (strong recommendation, based on **GOAL** in Ugandan children) advises **secondary prophylaxis for children with screen-detected RHD**, which reduced progression from 8.2% to 0.8%. A is wrong: untreated latent RHD can progress. B is wrong: she has no heart failure. C is wrong: mild MR does not need surgery. D is wrong: ASO monitoring does not prevent recurrence.

10. C · For 10 years or until age 25

C is correct: UCG 2023 recommends **10 years or until age 25** for carditis without residual disease (AHA: 10 years or age 21, whichever is longer). A is wrong: 5 years is for ARF without carditis. B is wrong: age 21 alone applies to the no-carditis category (UCG 18–21). D is wrong: lifelong is for residual valve disease. E is wrong: stopping early risks recurrence.

11. A · Take three sets of blood cultures before starting antibiotics, and arrange echocardiography

A is correct: fever, a new murmur, splinters, splenomegaly and haematuria in RHD suggest **infective endocarditis**; take **three blood cultures before antibiotics** and do echo, then give 4–6 weeks of IV antibiotics. B is wrong: ARF recurrence does not explain embolic phenomena. C is wrong: BPG prophylaxis does not treat endocarditis. D is wrong: a 3-week fever with these signs needs cultures; malaria testing can be added but is not the answer. E is wrong: steroids would worsen infection.

12. D · Reconsider the diagnosis (e.g. septic arthritis, JIA, sickle cell disease, leukaemia, post-streptococcal reactive arthritis)

D is correct: ARF arthritis typically responds **dramatically within 24–72 hours** to salicylates or NSAIDs; failure to respond should prompt reconsidering the diagnosis. A is wrong: doubling risks salicylate toxicity. B is wrong: steroids do not fix a wrong diagnosis and may hide leukaemia. C is wrong: waiting delays an alternative diagnosis such as septic arthritis. E is wrong: BPG prophylaxis is unrelated to the joint response.

13. C · An apical pansystolic murmur radiating to the axilla

C is correct: **mitral regurgitation** (apical pansystolic murmur radiating to the axilla) is the commonest finding in acute rheumatic carditis. A is wrong: that is a patent ductus arteriosus. B is wrong: a fixed split S2 suggests an atrial septal defect. D is wrong: that suggests pulmonary valve stenosis. E is wrong: a venous hum is an innocent murmur.

14. E · Use effective contraception (e.g. a copper IUD, implant or progestogen-only method), plan any pregnancy with the cardiologist and obstetrician, because warfarin is teratogenic in early pregnancy and mechanical valves carry high thrombosis risk

E is correct: mechanical valves need anticoagulation throughout pregnancy and pregnancy is high risk; **warfarin is teratogenic** (especially weeks 6–12), so pregnancy must be planned with specialists, and reliable non-oestrogen contraception is preferred. A is wrong: stopping anticoagulation risks valve thrombosis. B is wrong: risk is high. C is wrong: oestrogen increases thrombosis risk. D is wrong: warfarin embryopathy occurs in the first trimester.

15. B · In moderate/high-risk populations, an ESR ≥30 mm/h counts as a minor criterion

B is correct: high-risk thresholds are **ESR ≥30 mm/h and/or CRP ≥3.0 mg/dL**, with fever ≥38.0 °C. A is wrong: subclinical (echo-detected) carditis counts as major since 2015. C is wrong: PR prolongation cannot be counted when carditis is already a major criterion. D is wrong: chorea alone is enough. E is wrong: the fever threshold is ≥38.0 °C (high risk) or ≥38.5 °C (low risk).

SELF-TEST

Clinical cases

Work through each step before reading the model answer.

Case 1: The boarding-school girl with jumping joints

Atim, 12 years, a boarder at a primary school in Lira, is brought to the regional referral hospital with 10 days of painful joints and fever. She was treated at the school sick bay with paracetamol.

1. What history will help you most?

> **Pattern of joint involvement** (migratory, large joints, very painful), **sore throat 2–4 weeks ago**, skin sores, fever, chest pain/breathlessness/orthopnoea, abnormal movements or emotional change, rash/lumps, previous ARF or injections, **penicillin allergy**, family history, crowding, nutrition, immunisation, HIV status, menarche.

2. History: right knee then left ankle then right wrist, sore throat 3 weeks ago untreated; mild exertional breathlessness. Exam: 31 kg, T 38.4 °C, HR 120 (and 112 asleep), swollen tender left ankle, apical 3/6 pansystolic murmur to the axilla, no gallop, liver not enlarged. What criteria does she meet?

> **Major**: polyarthritis (migratory) and **carditis** (MR murmur, tachycardia out of proportion). **Minor**: fever ≥38.0 °C, plus ESR/CRP once available. With GAS evidence (ASO), this is **initial ARF with carditis**.

3. What investigations do you request and what do you expect?

> **ESR/CRP** (raised), **ASO** ± anti-DNase B (raised), **throat swab** (may be negative), **FBC**, **ECG** (PR may be long, but it cannot count as minor because carditis is major), **CXR** (heart size, congestion), **echo** (pathological MR per WHF criteria; check for MS, AR, effusion, LV function).

4. ESR 92, ASO 720, echo moderate MR with thickened leaflet tips, LV mildly dilated, good function. Write her management.

> Admit; rest while active then mobilise. **BPG 1.2 MU deep IM today** (≥30 kg) to eradicate GAS and start prophylaxis. **Aspirin 80–100 mg/kg/day in 3–4 doses** (≈2.5–3 g/day; max ~4 g) or **naproxen 10–20 mg/kg/day in 2 doses (max 1 g)** until joints settle and CRP falls, then taper. Monitor for heart failure (weight, RR, HR, liver). Consider enalapril if LV dilatation progresses. Register in the RHD clinic.

5. How long will she need BPG and what will you tell her family?

> **Every 4 weeks** for at least **10 years or until age 25** (UCG), and **until 40–45 or for life** if valve disease persists on follow-up echo. Explain that injections stop further attacks that damage the heart; missing them is dangerous; pain can be reduced; she must come early for breathlessness or swelling; good dental care; future pregnancies must be planned with a doctor.

6. What would make you suspect an alternative diagnosis during follow-up?

> **No response of arthritis within 72 hours** of salicylates/NSAIDs (septic arthritis, JIA, SCD, leukaemia), persistent fever with new murmur and embolic signs (**endocarditis**), cytopenias or organomegaly (leukaemia).

Take-away: Migratory polyarthritis with a new MR murmur after a sore throat is ARF with carditis. Apply high-risk Jones criteria, give BPG on day one, treat inflammation, echo everyone, and commit the family to years of 4-weekly injections.

Case 2: Breathless at 13: established RHD in heart failure

Okot, a 13-year-old boy (29 kg), is carried into the paediatric ward at night unable to lie flat. He was diagnosed with 'heart disease' 2 years ago and was supposed to get monthly injections but stopped after 4 because the health centre ran out.

1. What are your first actions?

> **Sit him up**, **oxygen** to SpO_2 >90%, IV access, vitals (HR, rhythm, BP, RR, SpO_2, temperature), **HemoCue Hb**, glucose. Look for **gallop, crackles, liver size, JVP, oedema**. Avoid IV fluid boluses.

2. Findings: RR 48, SpO_2 87%, HR 140 irregularly irregular, BP 96/58, gallop, liver 6 cm, bilateral crackles, apical PSM + mid-diastolic rumble with loud S1. Hb 10.5 g/dL, T 37.4 °C. What is going on?

> **Acute decompensated heart failure** from **chronic RHD with mixed mitral valve disease (MR + MS)**, precipitated by **new atrial fibrillation** and possibly **ARF recurrence** (prophylaxis stopped). Also consider endocarditis and infection.

3. Write the immediate treatment.

> **Oxygen**, sit up. **Furosemide 1 mg/kg IV (≈30 mg)**; if no diuresis in 2 h, **2 mg/kg**. **Rate control for AF**: **digoxin** oral load 15 µg/kg (WHO; ≈435 µg, often split into two doses 6 h apart) then 5 µg/kg 12-hourly (max 250 µg/dose), or a beta-blocker when not in acute failure. Check K^+ (avoid hypokalaemia with digoxin). Fluid and salt restriction. ECG, CXR, ESR/CRP, ASO, blood cultures if febrile. **Anticoagulation** for AF with MS (heparin then warfarin) after specialist discussion.

4. Investigations show ESR 70, ASO 560, CRP 48. Echo: moderate MS (mitral valve area reduced), moderate MR, left atrium dilated, pulmonary artery pressure raised. How does this change your plan?

> Likely **recurrent ARF** on chronic RHD. Restart **BPG every 4 weeks** (at 29 kg: **0.6 MU** by the UCG 30 kg cut-off; AHA would give 1.2 MU above 27 kg; follow your clinic protocol and re-weigh when euvolaemic), treat inflammation (aspirin or naproxen) if arthritis/active carditis, continue diuretics and rate control. Refer to the **Uganda Heart Institute** for assessment for **balloon mitral valvotomy or surgery**.

5. Why should you be cautious with ACE inhibitors here?

> In **stenosis-dominant** disease, cardiac output depends on filling across a narrowed valve; vasodilators can drop BP without relieving the obstruction. ACE inhibitors are most useful for **regurgitant** lesions. Rate control (longer diastole) and diuretics help MS; definitive treatment is mechanical.

6. How will you prevent this happening again?

> Register in an **RHD clinic**, provide a **BPG card** and schedule close to home, use pain-reduction methods, contact the community health worker for **tracing defaulters**, educate the family and school, report **BPG stock-outs** and arrange alternative supply (oral penicillin V 250 mg twice daily as a fallback), and plan follow-up echo and anticoagulation monitoring.

Take-away: Stopping prophylaxis leads to recurrence and decompensation. Treat failure and AF, restart BPG, refer stenosis for mechanical relief, and build adherence support around the family.

Case 3: The clumsy girl

Nansubuga, a 9-year-old girl (23 kg) from Mukono, is brought because her teacher reports she has become 'restless and careless', cannot write neatly, and laughs and cries easily. Her mother noticed she drops cups.

1. What examination findings will you look for?

> **Chorea**: involuntary purposeless movements of face and limbs, **milkmaid's grip**, **darting tongue**, **pronator sign**, **spooning**, hypotonia, difficulty with a spiral drawing; check the movements disappear in sleep. Full **cardiovascular exam** (murmurs, tachycardia), joints, skin (nodules, erythema marginatum), growth, BP.

2. She has milkmaid's grip and a positive pronator sign. Heart sounds normal, no murmur. ESR 20, ASO 150 IU/mL (normal). Can you diagnose ARF?

> **Yes**: **Sydenham chorea alone** diagnoses ARF, without other criteria or GAS evidence, because chorea appears 1–6 months after infection when ASO and inflammatory markers may have normalised. Exclude other causes (drugs, SLE, Wilson, hyperthyroidism) clinically.

3. What investigation is still essential?

> **Echocardiography**: subclinical carditis is common with chorea and determines the **duration of prophylaxis**. (Hers shows mild pathological MR: subclinical carditis.)

4. Write her treatment.

> **BPG 0.6 MU IM** (<30 kg) now and **every 4 weeks**. **Chorea**: calm environment, explain to school; if disabling, **sodium valproate 10–20 mg/kg/day** in 2–3 doses (≈115–230 mg 12-hourly) or carbamazepine 7–20 mg/kg/day. No aspirin needed (no arthritis). Prophylaxis duration: carditis present → at least **10 years or until 25** (UCG), longer if residual disease on follow-up echo.

5. How long will the chorea last and what should you tell the teacher?

> Usually **weeks to months** (typically resolves within 6 months, sometimes recurs). Explain it is a **medical illness, not misbehaviour**; allow extra time for written work, avoid punishment, and reduce stress.

6. Present her on the ward round in one paragraph.

> "Nansubuga, 9 years, 23 kg (WAZ −0.8), immunised per UNEPI, HIV-negative, presents with 6 weeks of involuntary movements, clumsiness and emotional lability; exam shows Sydenham chorea with no murmur; ASO and ESR normal; echo shows mild pathological MR (subclinical carditis). Diagnosis: ARF (Sydenham chorea) with subclinical carditis. Plan: BPG 0.6 MU 4-weekly for at least 10 years or until 25, valproate for disabling chorea, school liaison, repeat echo in 1 year."

Take-away: Chorea is enough to diagnose ARF even with normal ASO. Always echo, because subclinical carditis decides how long BPG must continue.

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A study aid, not a protocol. Check doses against the Uganda Clinical Guidelines 2023, your unit's protocol and a senior before you treat a patient.